



Gas Trading — Deep-Dive Q&A

Companion to the EU/US Gas Trading Handbook

Q1. When might the TTF forward curve NOT exhibit contango summer→winter / backwardation winter→summer?

The "natural" shape — contango from summer to winter, backwardation from winter to summer — reflects the **storage incentive cycle**: the market pays a premium for winter gas to incentivise injections during summer. But this shape breaks down in several real-world scenarios:

Scenario A: Acute Near-Term Supply Crisis (Curve Flips to Full Backwardation)

If there's a **current** supply emergency — say a major Norwegian pipeline outage in July, or a geopolitical shock cutting LNG flows — the **front of the curve spikes** while the back stays anchored. The market prices immediate scarcity, but assumes the crisis resolves over time. Result: the entire curve goes into **backwardation** (prompt > deferred), even from summer to winter.

Example: In August 2022, TTF front-month traded above €300/MWh while Calendar 2024 was at ~€50/MWh. The entire curve was in steep backwardation because the market was panicking about *current* supply, not future supply.

Scenario B: Oversupplied Winter / Storage Overhang (Flat or Inverted Winter)

If storage is **extremely full** heading into winter (say 95%+), the winter premium collapses. The market doesn't need to pay a scarcity premium for winter gas because there's plenty in storage. The winter contracts may even trade *at or below* summer levels.

Scenario C: Structural Demand Destruction

If industrial demand has permanently fallen (as it partially has post-2022), the winter heating peak is less pronounced relative to baseload, flattening the seasonal spread.

Scenario D: LNG Oversupply

A wave of new global LNG capacity can flood Europe with cheap gas year-round, compressing seasonal spreads because storage is no longer the binding constraint — you can get gas shipped in anytime.

The Key Principle

*The seasonal shape exists **because storage is the balancing mechanism**. Anything that changes the reliance on storage — abundant alternative supply, demand destruction, extreme tightness — distorts the natural shape.*

Q2. What is regas capacity and how is it relevant to the NBP?

What Is Regasification?

LNG (Liquefied Natural Gas) is natural gas that has been **cooled to -162°C** , turning it from gas into liquid for shipping on specialised tankers. When it arrives at its destination, it needs to be **heated back into gas** before it can enter the pipeline network. This process is called **regasification** ("regas").

A **regasification terminal** is the physical facility that: 1. Receives LNG tankers at a jetty 2. Stores LNG in cryogenic tanks 3. Warms the LNG back to gaseous form (using seawater heat exchangers or combustion) 4. Injects the gas into the national pipeline grid

Regas capacity = the maximum throughput rate of these terminals, measured in Bcm/year or GW.

Why It Matters for NBP

The UK has **disproportionately high regas capacity** relative to its gas demand, primarily through: - **Grain LNG** (Kent) — one of Europe's largest terminals - **South Hook LNG** (Milford Haven) — receives Qatari LNG - **Dragon LNG** (Milford Haven)

This means the UK can physically import **more LNG than it needs domestically**. When global LNG is abundant and cargoes are looking for a home, the UK absorbs the excess — flooding the UK market and pushing **NBP prices down** relative to TTF.

But here's the constraint: the UK can only **export** this surplus gas to continental Europe via two interconnector pipelines (IUK and BBL), which have **limited capacity**. So when the UK is saturated with LNG but can't push it all to Europe fast enough, NBP trades at a **discount** to TTF.

Conversely, when global LNG is tight and fewer cargoes arrive in the UK, the UK relies more on its own limited production and interconnector imports from Europe, and NBP can spike **above** TTF.

***In short:** The UK's large regas capacity makes NBP a "shock absorber" for global LNG supply swings, which is why NBP-TTF spread dynamics are fundamentally about how much LNG the UK is importing relative to what it can consume and re-export.*

Q3. What does it mean that the NBP-TTF spread is a trade on interconnector utilisation and UK LNG imports?

Let me break this into two parts:

Part 1: Interconnector Utilisation

The UK and continental Europe are connected by two physical gas pipelines: - **IUK (Interconnector):** Bacton (UK) ↔ Zeebrugge (Belgium) — bidirectional, ~20 GW - **BBL Pipeline:** Balgzand (Netherlands) → Bacton (UK) — originally one-way into the UK, now bidirectional

These are the **only** physical links between UK and EU gas markets. Their combined capacity is **limited**.

When you trade the NBP-TTF spread, you are fundamentally betting on whether these pipes are: - **Underutilised** (spare capacity) → Gas flows freely between UK and EU → Prices converge → Spread is tight - **At capacity** (fully utilised) → Markets decouple → Spread blows out

If the UK has excess gas (lots of LNG arriving) but the interconnectors are maxed out exporting to Europe, NBP drops below TTF. If Europe has excess gas but can't push enough through to the UK during a UK cold snap, NBP spikes above TTF.

Part 2: How LNG Imports Are Relevant

The UK's gas supply comes from three sources: 1. **North Sea / domestic production** (declining) 2. **Interconnector imports** from Europe (via IUK/BBL) 3. **LNG imports** (via Grain, South Hook, Dragon terminals)

LNG imports are the **swing variable**. They're not fixed — they depend on global LNG market conditions: - If JKM (Asian price) is high, LNG cargoes divert to Asia → fewer UK LNG arrivals → UK gets tighter → NBP rises → NBP-TTF spread widens (NBP premium) - If JKM is low and LNG is abundant globally, cargoes flood into UK terminals → UK is awash with gas → NBP falls → NBP-TTF spread widens (NBP discount)

***The trade:** If you think global LNG supply will be abundant next month (e.g., new Qatari trains coming online), you'd expect more cargoes into the UK → NBP weakens relative to TTF → **Short NBP, Long TTF** (sell the NBP-TTF spread).*

Q4. How are global LNG flows currently impacting the arbitrage bands between benchmarks?

The Three Key Benchmarks

Benchmark	Region	Current Price (~Sep 2026)
Henry Hub (HH)	US	~\$2.80–3.00/MMBtu
TTF	Europe	~€75–80/MWh (~\$25–27/MMBtu)
JKM	Asia	Typically tracking close to TTF ± \$1–2/MMBtu

Current Dynamics

The **transatlantic spread** (TTF minus HH) is extremely wide (~\$22–24/MMBtu). This is far above the total shipping cost (~\$4–5/MMBtu including liquefaction, freight, regas), meaning the **arbitrage is wide open** — every available US LNG cargo has a massive economic incentive to go to Europe (or Asia).

So why doesn't the spread close? Because:

1. **US LNG export capacity is physically maxed out.** The US can only liquefy and export ~17.4 Bcf/d. All terminals are running at near full capacity. You can't ship more LNG just because the arb is attractive — you

need physical terminal capacity.

2. **New capacity takes years to build.** Plaquemines LNG, Golden Pass, etc. are coming online gradually, but it's a multi-year buildout.
3. **The TTF-JKM spread is tight** (~\$1–2/MMBtu), meaning Europe and Asia are competing for the same marginal cargoes. When both basins are tight, the Atlantic arb stays wide because the constrained supply can't satisfy both.

Impact on Arbitrage Bands

- The **no-arbitrage band** (the spread level where it's not worth shipping) is ~\$4–5/MMBtu
- The **current spread** is ~\$22–24/MMBtu — far above the band
- This means the arbitrage is **open but physically constrained** — the price signal is screaming "send more gas to Europe" but the infrastructure can't respond fast enough
- The band only tightens when: (a) more LNG capacity comes online, (b) EU demand drops enough to reduce tightness, or (c) HH rises significantly

Q5. What are regas terminals? What is DES (Delivered Ex-Ship)?

Regas Terminals

As covered in Q2: these are physical facilities that receive LNG tankers, store the liquid in cryogenic tanks, warm it back into gaseous form, and inject it into the pipeline network. They are the "gateway" through which seaborne LNG enters a country's gas grid.

There are two types: - **Onshore terminals:** Permanent facilities built on land (e.g., Grain LNG in UK, Gate Terminal in Netherlands) - **FSRUs (Floating Storage Regasification Units):** Ships that can receive LNG and regasify it offshore, moored near a coast. Much faster and cheaper to deploy — Europe added many FSRUs post-2022 (Germany added 4+ FSRUs in record time).

DES (Delivered Ex-Ship)

DES is a **pricing/delivery term** from international trade (Incoterms). It means:

The seller delivers the LNG to the buyer's port, still on the ship. The seller bears all costs and risks of transportation up to the point of arrival at the destination port.

In gas trading, the **DES price** is the price of LNG **at the receiving port**, before regasification. It's what a buyer pays for the cargo "as delivered to the ship's rail."

Why DES ≠ TTF

Here's the critical distinction:

$$\text{TTF Price} = \text{DES Price} + \text{Regasification Cost} + \text{Pipeline Entry Tariff}$$

So:

$$\text{DES Price} = \text{TTF Price} - \text{Regas Cost} - \text{Tariffs}$$

When regas capacity is **congested** (all terminal slots are booked), the regas cost spikes (it's effectively an auction for scarce capacity). This means the DES price drops **below** TTF by a wide margin. The implication: even if TTF is very high, LNG sellers might not earn as much as TTF suggests because the regas fee eats into their margin.

Interview insight: If the interviewer asks "does a spike in TTF automatically attract more LNG cargoes?" — the answer is **not necessarily**. What matters is the DES netback, which depends on regas availability. A TTF spike with congested regas terminals doesn't translate to attractive LNG import economics.

Q6. Where can I find EU storage and terminal capacity? Is it publicly available?

Yes, absolutely — and you should be checking these regularly. Here are the key sources:

Storage Data

Source	URL	What It Shows
GIE AGSI+	agsi.gie.eu	THE primary source. Daily EU gas storage levels by country, facility, operator. Shows injection/withdrawal rates, % full, comparison to prior years. Free to access.
GIE ALSI	alsi.gie.eu	LNG terminal data — send-out, inventory, utilisation rates for all European LNG terminals.
EIA (US)	eia.gov	US weekly storage report.

Terminal Capacity Data

Source	What It Covers
GIE LNG Database	Comprehensive database of all European LNG terminals — nameplate capacity, operator, status, expansions planned
ENTSOG Transparency Platform	Pipeline flows, interconnector capacity, nominations — real-time data on physical gas flows across Europe
ACER REMIT	Transaction reporting data (more for compliance than trading)

How to Use AGSI+ in Practice

1. Go to agsi.gie.eu
2. Check the **EU aggregate** storage level (currently ~69%)
3. Compare to the **5-year range** — where are we relative to historical norms?
4. Look at the **injection/withdrawal rate** — is it faster or slower than seasonal norms?
5. Project forward: at the current rate, will we hit the **90% fill target** by 1 November?

Tip for interview: If asked "how do you track European gas fundamentals?", mention AGSI+ by name. It shows you know the tools that actual gas traders use daily.

Q7. What is the clean spark spread vs coal-gas dispatch? What is the clean spark spread exactly?

The Spark Spread (Basic)

The **spark spread** is the theoretical **gross profit margin** of a gas-fired power plant. It answers: "If I buy gas and burn it to generate electricity, what's my margin?"

$$P_{\text{Spark Spread}} = P_{\text{electricity}} - (P_{\text{gas}} \times \text{HR})$$

Where: - $P_{\text{electricity}}$ = wholesale electricity price (€/MWh) - P_{gas} = gas price (€/MWh thermal)
 - HR = **heat rate** = how much gas (in thermal MWh) you need to burn to produce 1 MWh of electrical output.
 Typical gas plant: HR = 1.8–2.5 (i.e., efficiency of 40–55%)

Example: Electricity at €90/MWh, gas at €35/MWh, heat rate 2.0: $P_{\text{Spark Spread}} = 90 - (35 \times 2.0) = 90 - 70 = €20/\text{MWh}$

This is the **gross** margin before carbon costs.

The Clean Spark Spread (CSS)

In Europe, gas-fired power plants must buy **EU Allowances (EUAs)** under the EU Emissions Trading System to cover their CO₂ emissions. The **clean** spark spread subtracts this carbon cost:

$$P_{\text{CSS}} = P_{\text{electricity}} - (P_{\text{gas}} \times \text{HR}) - (\text{EF}_{\text{gas}} \times P_{\text{carbon}})$$

Where: - EF_{gas} = emissions factor of natural gas ≈ 0.37 tonnes CO₂ per MWh thermal (gas is relatively clean) - P_{carbon} = EUA price (€/tonne CO₂), currently $\sim €65\text{--}75/\text{t}$

Example continued: EUA at €70/t: $P_{\text{CSS}} = 90 - (35 \times 2.0) - (0.37 \times 2.0 \times 70) = 90 - 70 - 51.8 = -€31.8/\text{MWh}$

Negative CSS → It's **unprofitable** to run the gas plant after paying for carbon. The plant would shut down.

Coal-Gas Dispatch & Switching

Power generators dispatch (turn on) their cheapest plants first. The **switching price** is the gas price at which a gas plant becomes cheaper than a coal plant. This depends on: - Coal price - Gas price - **Carbon price** (critical! Coal emits $\sim 2\times$ the CO₂ per MWh as gas)

Since coal has ~ 0.9 t CO₂/MWh thermal vs gas at ~ 0.37 t CO₂/MWh thermal, a **higher carbon price favours gas** over coal (because coal's carbon penalty is larger).

Why Carbon Cost Is Subtracted

The CSS subtracts carbon because it's a **real operating cost** — you literally have to buy EUA allowances for every tonne of CO₂ you emit. If you don't account for it, you're overstating the plant's actual profit margin. The "clean" just means "net of carbon costs."

Why Gas Traders Care

The CSS determines **how much gas gets burned for power generation** — one of the largest demand sectors. If the CSS is positive, more gas plants run → higher gas demand → bullish for gas prices. If the CSS is negative, gas plants shut down, reducing gas demand → bearish.

Q8. Why has there been significant demand destruction for industrial use since 2022? Is it Covid?

No, it's not Covid. Covid was 2020–2021 and caused a **temporary** demand dip. The 2022 demand destruction is a **separate, far more structural** event driven by the **European energy crisis**.

What Happened in 2022

1. **Russia invaded Ukraine** (February 2022)
2. In retaliation for Western sanctions, Russia progressively **cut gas supplies to Europe** throughout 2022
3. Russian pipeline gas had been ~40% of EU supply — this was being eliminated
4. TTF prices **skyrocketed** from ~€25/MWh (early 2021) to a peak of **€340/MWh** in August 2022
5. At these prices, many industrial processes became **economically unviable**

How It Destroyed Industrial Demand

Industries that use gas intensively include: - **Fertiliser/ammonia production** (gas is both fuel AND feedstock) - **Ceramics and glass manufacturing** - **Metals smelting (aluminium, zinc, steel)** - **Petrochemicals** - **Paper and pulp**

At €300/MWh gas prices, the cost of running these operations exceeded the price they could sell their products for. Many factories: - **Shut down temporarily** (mothballed) - **Relocated production** outside Europe (to the US, Middle East, or Asia where gas was 10x cheaper) - **Permanently closed** facilities that were marginal even before the crisis - **Switched to alternative fuels** or **invested in efficiency** to reduce gas dependence

Why It's Partially Permanent

Even though gas prices have come down from €340/MWh, they remain **structurally higher** than the pre-2022 average (~€15–25/MWh). At €75–80/MWh: - Some factories that shut down have **not reopened** — the capital has moved elsewhere - Companies that invested in efficiency or fuel-switching won't revert back - European industry has structurally **resized** to operate with less gas - This is why it's called **demand destruction** (permanent) rather than **demand reduction** (temporary)

***Estimated magnitude:** EU industrial gas demand fell by ~15-20% from 2021 to 2023, and roughly half of that is considered permanent structural loss.*

Q9. What is Bcm? What's the conversion relationship with TWh?

Bcm = Billion Cubic Metres

This is the standard volumetric unit for large-scale gas quantities, primarily used in **European** gas market discussions.

- 1 Bcm = 1,000,000,000 m³ of natural gas (at standard conditions: 15°C, 1 atm)

The Conversion

$$1 \text{ Bcm} \approx 10.55 \text{ TWh} \quad (\text{thermal energy content})$$

Or equivalently:

$$1 \text{ TWh} \approx 0.0948 \text{ Bcm}$$

Practical Context

Quantity	Bcm	TWh
EU total annual gas demand (~2025)	~350 Bcm	~3,700 TWh
EU storage capacity	~100 Bcm	~1,100 TWh
Norway annual supply to EU	~110 Bcm	~1,160 TWh
1 standard LNG cargo	~0.07 Bcm	~0.74 TWh

Other Common Conversions

Unit	Relationship
1 Bcm ≈	35.3 Bcf (billion cubic feet)
1 Bcm ≈	10.55 TWh
1 Bcm ≈	36.0 million MMBtu
1 TWh ≈	3.412 billion Btu

Why Two Units Exist

- **Europe** uses Bcm and TWh (metric system; prices in €/MWh)
- **US** uses Bcf and MMBtu (imperial system; prices in \$/MMBtu)

Being able to convert fluently between these is essential for cross-Atlantic gas trading.

Q10. How would you know how % filled EU storage tanks are? Where can you find this?

Primary Source: GIE AGSI+

Website: <https://agsi.gie.eu>

This is run by **GIE (Gas Infrastructure Europe)**, the industry association of European gas infrastructure operators. Every storage operator in Europe is required to report their data.

What AGSI+ Shows

Data Point	Detail
Fill level (%)	Current working gas volume ÷ total working gas capacity × 100
Working gas volume	How much gas is currently in storage (TWh or Bcm)
Injection/Withdrawal rate	How fast gas is being added or removed (GWh/day)
Country breakdown	Storage by country (Germany, Italy, Netherlands, France, Austria, etc.)
Facility-level data	Individual storage facilities (e.g., Rehden in Germany, Stogit in Italy)
Historical comparison	Current fill vs. 5-year min/max/average for the same date

How Traders Use It

1. **Daily check:** Is storage tracking above or below the 5-year average?
2. **Rate analysis:** Is injection pace sufficient to hit the 90% November target?
3. **Country focus:** Germany and Italy hold the most storage — watch these closely
4. **End-of-season (EOS) projection:** Based on current trajectory, where will storage be at the end of winter?
This drives the winter risk premium.

US Equivalent

For US storage, traders use the **EIA Weekly Natural Gas Storage Report** (Thursdays at 10:30 AM ET), which reports total working gas in underground storage in **Bcf**, broken down by region (East, Midwest, Mountain, Pacific, South Central with salt vs non-salt).

Q11. Given that most NG contracts are rolled over, is there a cost associated with doing so?

Yes, there is absolutely a cost (or benefit) — and understanding this is critical for a trading interview.

What "Rolling" Means

When your front-month contract approaches expiry, you: 1. **Close** (sell) your expiring position in the prompt month
2. **Open** (buy) an equivalent position in the next month

You are not taking delivery — you're simply shifting your exposure forward in time.

The Roll Cost/Benefit

The cost of rolling depends on the **shape of the forward curve** at the time you roll:

In Contango (Next Month > Current Month)

You sell the cheap expiring contract and buy the more expensive next-month contract.

$$P_{\text{Roll Cost}} = P_{\text{M2}} - P_{\text{M1}} > 0 \quad \text{(you PAY the contango)}$$

Example: You're long at \$3.00 (M1). M2 is at \$3.15. You sell M1 at \$3.00, buy M2 at \$3.15. You've lost \$0.15/MMBtu × 10,000 MMBtu = **\$1,500 per contract** just to maintain the same position.

This is called **negative roll yield** or "**paying the carry.**"

In Backwardation (Next Month < Current Month)

You sell the expensive expiring contract and buy the cheaper next-month contract.

$$P_{\text{Roll Benefit}} = P_{\text{M1}} - P_{\text{M2}} > 0 \quad \text{(you EARN the backwardation)}$$

Example: M1 at \$3.50, M2 at \$3.35. You sell M1 at \$3.50, buy M2 at \$3.35. You've gained \$0.15/MMBtu = **\$1,500 per contract.**

This is called **positive roll yield** or "**earning the carry.**"

Impact on Returns

Over time, roll costs/benefits can be **substantial** — sometimes larger than the directional price move. Many commodity ETFs and passive long strategies suffer from persistent **contango drag** (paying to roll every month in a contango market). This is why gas ETFs often underperform the spot price.

Key interview point: The position is not "unaffected" — rolling has a real P&L impact. A good trader considers the **total return** (price change + roll yield), not just the price change.

Q12. What is the Bcf/d unit?

Bcf/d = Billion Cubic Feet per Day

This is the standard **flow rate** unit for natural gas in the US market.

- **Bcf** = Billion Cubic Feet (volume)
- **/d** = per day (rate)

Scale Context

Measurement	Value
Total US gas production	~111 Bcf/d
Total US gas consumption	~90–95 Bcf/d
US LNG exports	~17.4 Bcf/d
A single large gas plant	~0.5–1.0 Bcf/d
EIA weekly storage build	Reported in total Bcf (e.g., 75 Bcf injection over the week $\approx$ 10.7 Bcf/d net injection rate)

Related Units

Unit	Meaning	Scale
MMcf/d	Million cubic feet per day	1/1000 of Bcf/d
Tcf	Trillion cubic feet	Annual consumption (~33 Tcf/year in US)
MMBtu	Million British Thermal Units	$\approx$ 1 Mcf (thousand cubic feet) of gas

Quick Conversion

$1 \text{ Bcf/d} \approx 10.3 \text{ Bcm/year}$

So US production of ~111 Bcf/d $\approx$ ~1,140 Bcm/year.

Q13. Why would the petchem/fertilizers industry be part of demand for US gas?

Natural gas is not just a **fuel** in these industries — it's a **chemical feedstock** (raw material input).

Fertilizers (Ammonia Production)

The Haber-Bosch process produces **ammonia (NH₃)**, the basis of most nitrogen fertilizers:



Where does the **hydrogen (H₂)** come from? **Natural gas (methane, CH₄)** via steam methane reforming:



Natural gas provides both: 1. **The hydrogen atoms** that become part of the ammonia molecule (feedstock) 2. **The heat energy** to drive the reactions (fuel)

Gas is typically 70–80% of the cost of producing ammonia. This is why US fertilizer production boomed with cheap shale gas — the US became one of the cheapest places in the world to make fertilizer.

Petrochemicals

Natural gas liquids (NGLs) extracted from natural gas — particularly **ethane** — are the primary feedstock for: - **Ethylene** (the world's most-produced chemical, used in plastics, packaging, etc.) - **Propylene** (polypropylene plastics) - **Methanol** (solvents, formaldehyde, acetic acid)

US **ethane crackers** (chemical plants that break ethane into ethylene) have expanded massively because cheap US gas makes US ethane far cheaper than Middle Eastern or European naphtha-based production.

Why This Matters for Gas Trading

- **Price sensitivity:** When gas prices rise, fertilizer and petrochemical production becomes unprofitable → these plants cut gas consumption → demand falls. This creates a **demand floor** — at a certain price, industrial shutdown reduces demand enough to prevent further price increases.
- **Export linkage:** US ammonia and methanol exports effectively "export" embedded gas demand, tying US gas prices to global agricultural and chemical markets.

Q14. Why are LNG export terminal outages paradoxically bearish for US domestic prices?

This is a classic interview question designed to test whether you understand the physical gas flow mechanics rather than just pattern-matching on "outage = bullish."

The Logic Chain

Normal state:

US Gas Production → [Some goes to domestic market] + [Some goes to LNG terminals → Shipped abroad]

When an LNG terminal goes offline:

US Gas Production → [ALL goes to domestic market] ← No LNG exports

Step by Step

1. The US produces ~111 Bcf/d of gas
2. Of this, ~17.4 Bcf/d is consumed by LNG export terminals (liquefied and shipped overseas)
3. If a major terminal (say Freeport LNG at ~2.1 Bcf/d) shuts down due to an explosion, maintenance, or hurricane...
4. That 2.1 Bcf/d of gas that **was leaving the country** now has **nowhere to go**
5. It backs up into the domestic pipeline system
6. Domestic supply effectively increases by 2.1 Bcf/d
7. More supply with unchanged domestic demand → **prices fall**

Real Example: Freeport LNG (June 2022)

On 8 June 2022, Freeport LNG had an explosion and shut down for ~6 months. Henry Hub prices fell ~10% in the days following, despite the fact that an "outage" occurred — because the ~2 Bcf/d of gas that was being exported was suddenly trapped domestically.

The Interview Answer

*"An LNG terminal outage is **bullish for international prices** (TTF/JKM lose supply) but **bearish for domestic US prices** (Henry Hub gains supply). The gas that was being exported is now trapped in the US domestic market, increasing effective supply. This widens the HH-TTF spread. It's a reminder that LNG terminals are effectively a **demand centre** for US gas — when they go offline, demand falls."*

The Nuance (Bonus Points)

The bearish effect on HH is **temporary and front-loaded** — producers eventually respond by curtailing output if prices stay low. But in the near term, the supply glut is immediate and unambiguous.

Q15. Does the weekly EIA NatGas report only cover US storage, or international?

US only.

The EIA Weekly Natural Gas Storage Report covers **US underground storage** exclusively, broken down into 5 regions: 1. East 2. Midwest 3. Mountain 4. Pacific 5. South Central (further split into **salt** vs **non-salt** caverns — this matters because salt caverns can cycle gas much faster)

International Equivalents

Region	Report	Source	Frequency
US	Weekly Natural Gas Storage Report	EIA	Weekly (Thursday 10:30 AM ET)
EU	AGSI+	GIE	Daily
UK	Included in AGSI+ (plus Ofgem data)	GIE / Ofgem	Daily
Japan	LNG inventory data	METI	Monthly

Why Only US Matters for the EIA Report

The EIA report is specifically about the US market and the US gas balance. However, because the US is the world's largest gas producer and a major LNG exporter, this data has **global implications** — a bearish US storage surprise (big build) can weaken TTF too if it signals global oversupply or reduced LNG export incentive.

Q16. What's the cost of carry framework?

Definition

The **cost of carry** is the total cost of holding (carrying) a physical commodity or financial position over time. For natural gas futures, it determines the **theoretical relationship** between spot prices and futures prices.

The Formula

$$F(t,T) = S(t) \cdot e^{(r + c - y)(T-t)}$$

Where: - $F(t,T)$ = futures price for delivery at time T - $S(t)$ = current spot price - r = risk-free interest rate (financing cost of tying up capital) - c = **storage cost** (renting underground cavern space, insurance, maintenance) - y = **convenience yield** (the implicit benefit of holding physical gas *now*) - $(T-t)$ = time to delivery

Interpretation

The futures price should equal the spot price **plus** the cost of storing the gas until delivery, **plus** financing costs, **minus** any convenience yield.

- If $r + c > y$: **Contango** — futures > spot (it costs more to store than the benefit of having gas now)
- If $y > r + c$: **Backwardation** — futures < spot (the benefit of having gas now exceeds storage costs)

Why It Matters

The cost of carry sets the **fair value** for calendar spreads. If the market-implied carry (the actual spread between two months) exceeds physical storage costs, you can: 1. Buy gas now (spot) 2. Store it 3. Sell it forward (futures) 4.

Lock in a risk-free profit (cash-and-carry arbitrage)

This arbitrage keeps spreads anchored to physical storage costs — which is why calendar spreads are typically mean-reverting.

Q17. What exactly is "carry"? What does "pay to hold" or "earn from the roll" mean?

"Carry" = The Return from Simply Holding a Position Over Time

Think of it like this: if you buy a futures contract and do nothing, do you make or lose money just from the passage of time (independent of whether the outright price moves)?

"Pay to Hold" (Negative Carry — Contango)

Imagine you're **long** the front-month gas contract in a contango market ($M1 = \$3.00$, $M2 = \$3.15$):

- Each month, you must **roll**: sell $M1$, buy $M2$
- Every roll costs you $\$0.15/\text{MMBtu}$
- You're literally **paying** to maintain your long position

- It's like paying rent to stay in the same apartment

Analogy: Imagine you own a physical barrel of gas. Each month you pay the storage operator \\$.15 to keep it in the ground. That's what contango drag feels like.

"Earn from the Roll" (Positive Carry — Backwardation)

Now imagine you're **short** the front-month in a contango market (or **long** in backwardation, $M1 = \$3.15$, $M2 = \$3.00$):

- Each roll: sell the expensive M1, buy the cheaper M2
- Every roll earns you \\$.15/MMBtu
- You're being **paid** to hold your position
- It's like getting paid interest on a deposit

Why Traders Care Deeply About This

In relative value trading, you want to **be on the right side of the carry**:

*"I want to put on a trade where my **fundamental view** AND **carry** are aligned. If I think winter gas is going higher AND I earn positive carry waiting for that move, I'm being paid while I wait for my thesis to play out. If I have to pay negative carry, my thesis needs to be strong enough to overcome that cost."*

This is why experienced traders often say: "**The carry is your friend**" — or your enemy if you're on the wrong side.

Q18. Why is carbon cost subtracted from the Clean Spark Spread?

Because it's a **real, unavoidable operating cost** for any gas-fired power plant in Europe.

The Mechanism

Under the **EU Emissions Trading System (EU ETS)**, every power plant must: 1. **Measure** its CO₂ emissions 2. **Surrender** one EU Allowance (EUA) for every tonne of CO₂ emitted 3. **Buy** EUAs on the market if it doesn't have enough

A gas-fired power plant emitting 0.37 tonnes CO₂ per MWh of thermal input, with a heat rate of 2.0, and EUAs at €70/tonne:

$$\text{\text{\$}\text{Carbon cost per MWh electricity}} = 0.37 \times 2.0 \times 70 = \text{\text{\$}51.80\text{\text{MWh}}_e\text{\text{\$}}$$

This is a **cash cost** — the plant operator must physically buy and surrender these allowances. If you don't subtract it, you're overstating profitability. A plant might look profitable on a "dirty" spark spread but actually be **losing money** after carbon costs.

Why "Clean" Matters More Than "Dirty"

- The **dirty spark spread** (without carbon) tells you the thermodynamic margin

- The **clean spark spread** (with carbon) tells you the **economic** margin — whether the plant should actually run
- Power plant dispatch decisions are based on CSS, not the dirty spread
- Therefore, **gas demand from the power sector** depends on the CSS

Q19. How do you calculate the spark spread?

Step-by-Step Calculation

Given: - Electricity price: €95/MWh - Gas price: €35/MWh (thermal) - Plant efficiency: 50% (meaning heat rate = $1/0.50 = 2.0$) - CO₂ emissions factor for gas: 0.37 t CO₂ / MWh thermal - EUA carbon price: €70/t CO₂

Step 1: Dirty Spark Spread

$$\text{DSS} = P_{\text{elec}} - P_{\text{gas}} \times \text{HR} = 95 - 35 \times 2.0 = 95 - 70 = €25/\text{MWh}$$

Step 2: Carbon Cost

$$\text{Carbon Cost} = EF_{\text{gas}} \times \text{HR} \times P_{\text{carbon}} = 0.37 \times 2.0 \times 70 = €51.80/\text{MWh}$$

Step 3: Clean Spark Spread

$$\text{CSS} = \text{DSS} - \text{Carbon Cost} = 25 - 51.80 = -€26.80/\text{MWh}$$

Interpretation: This plant is **unprofitable** — it loses €26.80 for every MWh of electricity it generates. It should **not run**. This means less gas is burned for power → lower gas demand.

When the Plant Runs

For CSS > 0:

$$P_{\text{elec}} > P_{\text{gas}} \times \text{HR} + EF_{\text{gas}} \times \text{HR} \times P_{\text{carbon}}$$

$$P_{\text{elec}} > \text{HR} \times (P_{\text{gas}} + EF_{\text{gas}} \times P_{\text{carbon}})$$

$$P_{\text{elec}} > 2.0 \times (35 + 0.37 \times 70) = 2.0 \times (35 + 25.9) = 2.0 \times 60.9 = €121.80/\text{MWh}$$

Electricity would need to be above ~€122/MWh for this plant to be profitable. Below that, gas demand from power generation falls.

Q20. How exactly would Bayesian updating be used to model a surprise in the EIA report?

The Setup

Before the report (your prior): - You have a forecast/model that predicts end-of-season (EOS) storage - Based on current trends, your model says EOS storage will be **3,200 Bcf** with some uncertainty - You represent this as a **prior distribution**: $\text{EOS} \sim N(\mu_0 = 3200, \sigma_0^2 = 150^2)$

The surprise: - Consensus expected a **+75 Bcf** weekly injection - Actual report: **+55 Bcf** (a **-20 Bcf miss**, bullish)

Step 1: Interpret the Signal

The 20 Bcf miss in one week tells you something about the underlying supply-demand balance. If demand is running hotter than expected (or supply lower), this likely continues for multiple weeks.

You estimate that a -20 Bcf weekly miss implies **~100 Bcf lower EOS** (because the conditions causing the miss — e.g., higher-than-expected power burn — will persist for several more weeks).

This is your **likelihood / new data signal**: $\text{EOS}_{\text{data}} \sim N(\mu_{\text{data}} = 3100, \sigma_{\text{data}}^2 = 200^2)$

Step 2: Bayesian Update

For normal distributions, the posterior is also normal:

$$\mu_{\text{posterior}} = \frac{\mu_0 / \sigma_0^2 + \mu_{\text{data}} / \sigma_{\text{data}}^2}{1 / \sigma_0^2 + 1 / \sigma_{\text{data}}^2}$$

$$\sigma_{\text{posterior}}^2 = \frac{1}{1 / \sigma_0^2 + 1 / \sigma_{\text{data}}^2}$$

Plugging in:

$$\mu_{\text{post}} = \frac{3200 / 150^2 + 3100 / 200^2}{1 / 150^2 + 1 / 200^2} = \frac{3200 / 22500 + 3100 / 40000}{1 / 22500 + 1 / 40000}$$

$$= \frac{0.1422 + 0.0775}{0.0000444 + 0.0000250} = \frac{0.2197}{0.0000694} \approx 3165 \text{ Bcf}$$

$$\sigma_{\text{post}} = \sqrt{\frac{1}{0.0000694}} \approx 120 \text{ Bcf}$$

Step 3: Trading Decision

Before the report: You expected EOS = 3,200 Bcf. Market was priced accordingly.

After the report: Your updated estimate is EOS = 3,165 Bcf, with tighter confidence.

Action: The market is now mispriced (still reflecting ~3,200). You: - Go **long** winter gas (lower EOS = tighter supply = higher winter prices) - Or widen your winter-summer spread position - Size the trade proportional to how much your posterior shifted (35 Bcf shift = moderate conviction)

Step 4: Iterate Weekly

Each subsequent EIA report gives you new data. You update the posterior again, each time refining your EOS estimate. Over weeks, your distribution narrows and your conviction in the trade grows or shrinks.

Interview key: The power of Bayesian thinking is that it gives you a **principled framework** for incorporating new information without overreacting (anchoring too strongly on one data point) or underreacting (ignoring the signal).

Q21. Why are OU / Schwartz one-factor mean-reverting models more appropriate than GBM?

What GBM Assumes

Geometric Brownian Motion (the model behind Black-Scholes):

$$dS = \mu S dt + \sigma S dW$$

- Prices have a **drift** (trend) but **no tendency to return to any level**
- If the price doubles, GBM says it's equally likely to double again or halve
- Prices can wander to infinity or zero
- No concept of a "fair value" or "equilibrium"

Why That's Wrong for Gas

Gas prices have a **structural equilibrium** determined by physical fundamentals: - The **marginal cost of production** (if prices drop far below, producers shut wells → supply falls → price recovers) - The **demand destruction level** (if prices rise far above, industry shuts down → demand falls → price drops) - **Storage** acts as a physical buffer, smoothing out short-term imbalances

In other words, gas prices are "tethered" to a band defined by physical economics. **They mean-revert.**

What OU Captures

The Ornstein-Uhlenbeck process:

$$dX = \theta(\mu - X)dt + \sigma dW$$

- When price X is **above** μ : the drift term $\theta(\mu - X)$ is **negative** → pushes price down
- When price X is **below** μ : the drift term is **positive** → pushes price up
- θ controls how fast it reverts (higher = faster snap-back)
- The price **oscillates around** μ rather than wandering aimlessly

Empirical Evidence

Property	GBM Prediction	Gas Reality	OU Prediction
Autocorrelation of returns	Zero	Negative (reversals)	Negative ✓
Long-term price range	Unbounded	Bounded (~\$1–\$10 for HH)	Bounded ✓
Response to spike	Stays at new level	Reverts within weeks/months	Reverts ✓
Variance over time	Grows linearly with time	Bounded (converges)	Bounded ✓

Schwartz One-Factor

The Schwartz (1997) one-factor model is essentially OU applied to the **log** of commodity prices:

$$d\ln(S) = \theta(\mu - \ln S)dt + \sigma dW$$

This ensures prices stay positive (since you're mean-reverting the log-price) while still capturing mean reversion. It's the workhorse model for commodity derivatives pricing.

Q22. How does PCA work in principle? How does it segregate the 3 different "models"?

The Intuition

Imagine you have a gas forward curve with 36 monthly prices (M1 to M36). Every day, all 36 prices change. But they don't change independently — when the front of the curve moves up, the rest usually does too.

PCA asks: "What are the underlying driving forces that explain these 36 correlated movements?"

Step by Step

Step 1: Compute daily changes. Take the difference in each of the 36 monthly prices from one day to the next. You now have a matrix: rows = days, columns = tenors.

Step 2: Compute the covariance matrix. This 36×36 matrix captures how each tenor co-moves with every other tenor. The diagonal is the variance of each tenor; off-diagonal is the covariance between pairs.

Step 3: Eigendecomposition. Mathematically decompose this covariance matrix into **eigenvectors** (directions) and **eigenvalues** (magnitudes). Each eigenvector represents an independent "mode" of curve movement. Each eigenvalue tells you how much variance that mode explains.

Step 4: Rank by importance. Sort eigenvectors by their eigenvalues (largest first). The first eigenvector explains the most variance, the second explains the most remaining variance, and so on.

The Three Factors

For gas curves (and most commodity curves), the first 3 eigenvectors have intuitive interpretations:

PC	Interpretation	What It Looks Like	Variance Explained
PC1	Level / Parallel Shift	All tenors move by similar amounts in the same direction	~70–80%
PC2	Slope / Tilt	Front of curve moves opposite to back (steepening/flattening)	~10–15%
PC3	Curvature / Butterfly	Front and back move together, middle moves opposite (bowing)	~3–5%

They're Not Separate "Models"

PCA doesn't fit three separate models — it extracts three **orthogonal directions of maximum variance** from a single dataset. Think of it like this:

- Any daily curve change can be decomposed as: **"The whole curve shifted up by X, the front steepened by Y, and the belly bowed by Z"**
- Instead of tracking 36 correlated numbers, you track 3 independent factors

- This is **dimensionality reduction** — going from a 36-dimensional problem to a 3-dimensional one while retaining 95%+ of the information

Q23. What are eigenvectors, how do they tie into PCA, and what's the objective?

Eigenvectors and Eigenvalues — Plain English

For any square matrix A (like a covariance matrix), an **eigenvector** v is a special direction such that when you multiply the matrix by it, the result is just a **scaled version** of itself:

$$Av = \lambda v$$

- v = eigenvector (a direction in the data space)
- λ = eigenvalue (how much the data stretches in that direction)

Analogy

Imagine you have a cloud of data points (36-dimensional, one for each tenor). This cloud has a shape — it's elongated in some directions and compressed in others.

- The **eigenvectors** are the **principal axes of the ellipsoid** — the directions along which the data cloud is stretched the most
- The **eigenvalues** are the **lengths of those axes** — how much spread (variance) there is in each direction

How They Tie Into PCA

PCA = finding the eigenvectors of the covariance matrix, ranked by eigenvalue:

1. **1st eigenvector** (largest eigenvalue): The direction of **maximum variance** in the data. For gas curves, this is the "everything moves together" direction (level shift).
2. **2nd eigenvector** (2nd largest eigenvalue): The direction of maximum **remaining** variance, **orthogonal** (perpendicular) to the first. For gas curves, this is steepening/flattening.
3. **3rd eigenvector**: Maximum remaining variance orthogonal to both. For gas curves, this is curvature.

The Objective

The objective of PCA is to find a **low-dimensional representation** that captures as much variance as possible:

- **36 monthly prices** → **3 factors** that explain ~95% of movements
- This simplifies: risk management, hedging, signal generation, and understanding what's driving the curve
- It's NOT about "capturing more variance" for its own sake — it's about **parsimony**: explaining the complex reality of curve movements with the fewest possible independent drivers

Practical Use

Instead of monitoring 36 individual prices, a trader can think:

"Today the curve moved because PC1 (level) went up 2σ on the EIA surprise, and PC2 (slope) steepened 1σ because the surprise was front-loaded. PC3 (curvature) didn't move much."

This is far more actionable than saying "M1 went up \\$.012, M2 went up \\$.011, M3 went up \\$.09..." for all 36 contracts.

Q24. VaR: Is it multidimensional/subjective? How do firms standardise?

Yes, VaR Depends on Your Choices

VaR requires **three inputs**, and changing any of them changes the number:

Input	Typical Choices	Effect
Confidence level	95%, 99%, 99.5%	Higher confidence → larger VaR number
Time horizon	1 day, 10 days, 1 month	Longer horizon → larger VaR
Methodology	Historical, parametric, Monte Carlo	Different assumptions → different results

So yes, there is no single "VaR" — it's a function of these parameters. Two firms can report **different VaR numbers for the same portfolio** if they use different parameters.

Is VaR "Subjective"?

Partially, but that's by design. VaR is a **risk communication tool**, not a physical constant. The subjectivity comes from:

1. **Choice of confidence level:** 99% VaR ignores the worst 1% of scenarios. 95% VaR is less conservative.
2. **Choice of lookback period:** Historical VaR using 1 year of data gives different results than using 5 years.
3. **Distribution assumptions:** Parametric VaR assumes normality (underestimates tail risk); Historical doesn't.

How Firms Standardise

Despite the flexibility, the industry has converged on standards:

Context	Standard
Basel III / Bank regulation	99% confidence, 10-day horizon, at least 1 year of historical data
Internal risk management (most hedge funds, including Citadel)	99% 1-day VaR, often supplemented with 95% VaR
Reporting to investors	Usually 95% or 99% 1-day VaR

How Citadel Likely Uses VaR

Sophisticated firms like Citadel don't rely on VaR alone. They use a **risk framework stack**:

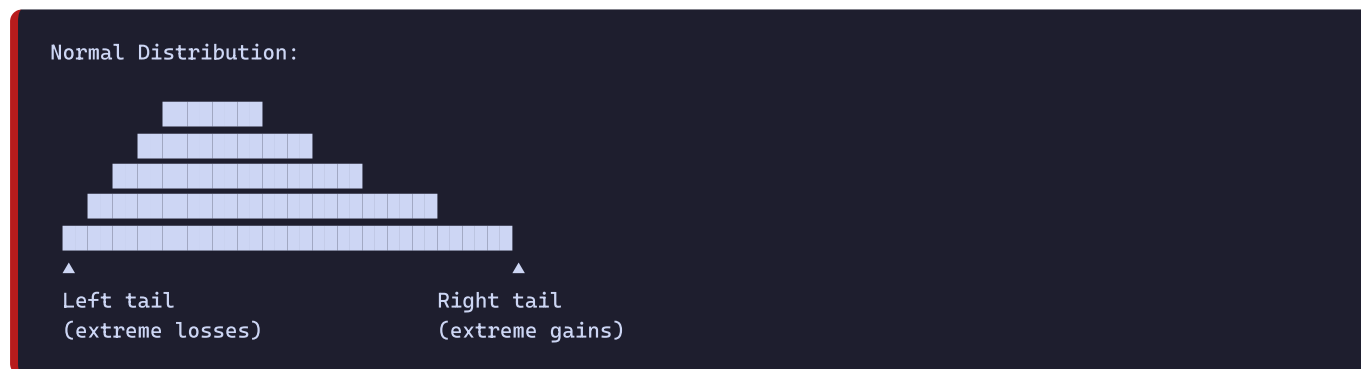
1. **VaR** (99%, 1-day) — daily risk limit
2. **Expected Shortfall / CVaR** — captures tail risk beyond VaR
3. **Stress tests** — historical scenarios (Uri, 2022 crisis) applied to current portfolio
4. **Concentration limits** — maximum position size per contract/spread
5. **Greeks limits** — delta, gamma, vega budgets
6. **Drawdown limits** — maximum cumulative loss before forced de-risking

Interview answer: "VaR is a useful starting point but has well-known limitations — it's not subadditive, it ignores tail shape, and it's parameter-dependent. That's why sophisticated desks supplement it with Expected Shortfall, stress testing, and scenario analysis."

Q25. What is "tail risk"? What does "gas has tail risk" and "kurtosis" mean?

Tail Risk — Definition

The **"tails"** of a probability distribution are the extreme ends — the very unlikely but very impactful outcomes. **Tail risk** is the risk of these extreme events occurring.



"Gas Has Tail Risk" — What This Means

Gas prices have a tendency to experience **extreme moves** far more frequently than a normal distribution would predict.

Under a normal distribution: - A 3σ event should happen ~0.3% of the time (once every 1.5 years) - A 5σ event should happen ~0.00006% of the time (once every 14,000 years)

In gas markets: - 3σ events happen several times per year - $5\sigma+$ events happen every few years (Winter Storm Uri was arguably $10\sigma+$)

Kurtosis — The Technical Measure

Kurtosis measures how "fat" the tails of a distribution are compared to a normal distribution:

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

- **Normal distribution:** Kurtosis = 3 (sometimes reported as "excess kurtosis" = 0)
- **Fat-tailed (leptokurtic):** Kurtosis > 3 → more extreme events than normal
- **Thin-tailed (platykurtic):** Kurtosis < 3 → fewer extreme events

Natural gas returns typically have kurtosis of 8–15+, meaning the tails are **dramatically fatter** than normal.

Why Gas Has Fat Tails

Gas has unique physical characteristics that create tail risk:

Driver	Mechanism
Weather extremes	Polar vortex events are non-linear demand shocks — a 5°F drop below –10°F has exponentially more impact than a 5°F drop from 40°F
Supply disruptions	Pipeline explosions, freeze-offs, LNG terminal outages remove supply discontinuously (sudden jumps, not gradual)
Storage constraints	When storage hits 0% or 100%, the market loses its buffer and prices become unanchored
Positive skew	Prices can spike to 10x their normal level but can't go below zero (or not far) — creating asymmetric tails

Why This Matters for Risk Management

If you use a **normal distribution** to calculate VaR, you'll dramatically underestimate the probability and magnitude of extreme losses. This is why: - Gas desks use **Student-t distributions** or **historical simulation** instead of Gaussian assumptions - **Expected Shortfall** (average loss in the tail) is preferred over VaR - **Stress testing** with actual historical extremes is essential

Q26. What is the H/J spread?

Contract Month Codes

Futures contracts use single-letter codes for delivery months:

Letter	Month	Letter	Month
F	January	N	July
G	February	Q	August
H	March	U	September
J	April	V	October
K	May	X	November
M	June	Z	December

H/J = March/April

The **H/J spread** is the March/April calendar spread: - **Long H/J** = Long March, Short April - **Short H/J** = Short March, Long April

Why It's Called the "Widow-Maker"

March (H) is the last month of winter — the final withdrawal month from storage. **April (J)** is the first month of spring — the start of injection season. The spread captures the **entire winter-to-spring transition**:

- If winter ends with **dangerously low storage** → March gas is scarce, March price spikes → H/J blows out (widens)
- If winter ends **comfortably** → No scarcity in March, March price drops → H/J collapses

The P&L swings can be enormous because late-winter weather is inherently unpredictable, and the physical consequences (running out of gas vs. having plenty) are binary.

Q27. What physical assets does Citadel own for its gas infrastructure?

Known Physical Assets

Citadel's physical gas operations are primarily through **Citadel Energy Marketing (CEM)**, and they have built a notable physical infrastructure footprint:

Asset Type	Details
Upstream Production	Apex Natural Gas — Citadel-backed gas production company. Gives CEM direct access to wellhead gas and production data before it hits the public market.
Pipeline Capacity	CEM holds firm transportation (FT) capacity on major US pipeline systems. This gives them the right to physically move gas between regions, enabling location arbitrage. They reportedly hold capacity on pipes connecting the Permian, Gulf Coast, and Northeast.
Storage Capacity	CEM leases underground storage capacity in multiple US regions. This allows them to inject gas when cheap and withdraw when expensive — effectively monetising the calendar spread physically.

Why Physical Assets Matter for a Financial Trader

This is a critical point about Citadel's edge:

1. **Informational advantage:** When you're actually moving physical gas, you see real-time flow data, nominations, and pipeline conditions *before* they show up in public data. You know when a pipeline is constrained because *your own gas isn't flowing*.
2. **Optionality:** Physical storage and transport are like **real options**: - Storage = a calendar spread option (inject when contango, withdraw when backwardation) - Pipeline capacity = a locational spread option (ship gas from cheap to expensive hubs) - These options have value that's not visible in a purely financial book

3. **Market-making ability:** Physical positions allow CEM to make markets in basis and storage — providing liquidity while earning the bid-ask spread.
4. **Fundamental intelligence:** Owning production via Apex gives Citadel direct insight into well economics, decline curves, and drilling activity — leading indicators for future supply.

Scale

At its peak, CEM was trading volumes equivalent to approximately **11% of total US natural gas consumption** — making it one of the largest physical gas traders in North America, comparable to major utilities and traditional commodity houses.

Interview implication: When talking to the hiring manager, acknowledge that Citadel's gas desk is not a pure "paper trading" operation. Their physical footprint is a key competitive advantage, and understanding how physical and financial markets interact is essential for the role. If asked "what's Citadel's edge in gas?", physical infrastructure and data advantage should be part of your answer.

Study priority for Tuesday: Focus especially on Q7/Q18/Q19 (spark spread — shows you understand gas demand drivers), Q14 (the LNG terminal trick question), Q17 (carry — fundamental to spread trading), Q20 (Bayesian updating — shows quant thinking), and Q27 (shows you've researched Citadel specifically). Good luck!

